

Name and Student ID: _____

1. (40 points) For each of the following questions choose **one answer** from A to D.

- (a) Let $f(x, y) = \sin(xy)/(x^2 + y^2)$ be a function defined on the plane $\mathbb{R}^2$ except the origin. Which of the following statements is true about $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$?

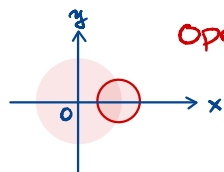
A. The limit is 0 because $f(0, y) = 0$ for all $y \neq 0$.
 B. The limit does not exist because f is not defined at $(0, 0)$. *← Doesn't matter.*
☒ C. The limit does not exist because f approaches different values as (x, y) approaches $(0, 0)$ from different directions.
 D. None of the above is true.

*If we set $y = kx$ and take $x \rightarrow 0$,
 $f(x, kx) = \frac{\sin(kx^2)}{(1+k^2)x^2} \rightarrow \frac{k}{1+k^2}$. This depends on k .*

- (b) Which of the following statements is true about the region in the plane $\mathbb{R}^2$ defined as the union of the set $\{(x, y) \mid x^2 + y^2 < 2\}$ and the set $\{(x, y) \mid (x-2)^2 + y^2 \leq 1\}$?

A. The region is open but not closed.
 B. The region is closed but not open.
 C. The region is open and closed.
☒ D. The region is neither open nor closed.

*Open disk $\cup$ closed disk
 Contains some boundary points but not all of them.*



- (c) Which of the following statements is true about the function $f(x, y) = \cos(xy)$?

A. $f_{xy}(1, \pi) = 0$.
☒ B. $f_{xy}(1, \pi) = \pi$.
 C. $f_{xy}(1, \pi) = -\pi$.
 D. $f_{xy}(1, \pi)$ is undefined.

*$f_x = -y \sin(xy)$
 $f_{xy} = -\sin(xy) - xy \cos(xy)$
 $f_{xy}(1, \pi) = \pi$*

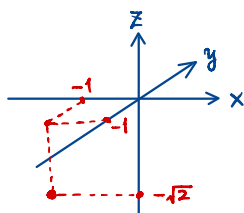
- (d) What is the union of all contour curves of the function $f(x, y) = \sqrt{1 - x^2 - y^2}$ defined in the region where $x^2 + y^2 \leq 1$?

A. A disk
 B. A sphere
☒ C. A hemisphere
 D. None of the above

Union of all contour curves is the surface $z = f(x, y)$.

- (e) What is the spherical coordinates (ρ, ϕ, θ) of the point with Cartesian coordinates $(x, y, z) = (-1, -1, -\sqrt{2})$?

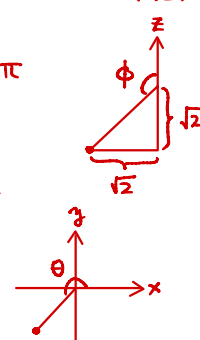
A. $(2, \pi/4, \pi/4)$
 B. $(2, 3\pi/4, \pi/4)$
 C. $(2, \pi/4, 5\pi/4)$
☒ D. None of the above



$$\rho = \sqrt{(-1)^2 + (-1)^2 + (-\sqrt{2})^2} = 2$$

$$\phi = \frac{3}{4}\pi$$

$$\theta = \frac{5}{4}\pi$$



- (f) Coordinates (u, v, w) are related to the Cartesian coordinates (x, y, z) by $x = u \cos(v)$, $y = u \sin(v)$, $z = w$. What is the Jacobian $\partial(x, y, z)/\partial(u, v, w)$?

☒ A. u
☐ B. v
☐ C. w
☐ D. None of the above

(u, v, w) are really the cylindrical coordinates (r, θ, z) .

$$dx dy dz = r dr d\theta dz$$

$$\Rightarrow \text{Jacobian} = r = u.$$

- (g) A two-dimensional object has the shape of a disk of radius 1. Its density function δ is unknown, but we know that the double integral of δ over the object is π . What is the mass of the object?

☐ A. 1
☒ B. π
☐ C. π^2
☐ D. None of the above

$$= \text{mass}$$

- (h) Which of the following statements is not true about the standard linear approximation of a function $f(x, y)$ at (a, b) ?

☐ A. It is equivalent to considering Taylor's formula for $f(x, y)$ about (a, b) to first order.
☐ B. It is equivalent to considering the tangent plane to the graph $z = f(x, y)$ through the point $(a, b, f(a, b))$.
☐ C. For some choices of f , this approximation gives an exact (that is, not approximate) answer.

☒ D. All of the above are true.

$\nwarrow$ If f is a linear in x and y , the linear approximation gives f .

- (i) What is the double integral of $x^2 \sin(x + y)$ over the region defined by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$?

☒ A. 0
☐ B. 1
☐ C. -1
☐ D. None of the above

Contributions from the neighborhood of (x, y) and the neighborhood of $(-x, -y)$ cancel.

- (j) A conservative vector field $\mathbf{F}$ in space has a potential function $f(x, y, z) = x^2 - y^2 - xz + z^2$. What is the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the helix C given by $\mathbf{r}(t) = \cos(2\pi t)\mathbf{i} + \sin(2\pi t)\mathbf{j} + t\mathbf{k}$, where the parameter t varies from 0 to 1?

☒ A. 0
☐ B. $\pi/2$
☐ C. $\sqrt{3}\pi$
☐ D. None of the above

$$\begin{aligned} \mathbf{F} = \nabla f &\Rightarrow \int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) \\ &= f(1, 0, 1) - f(1, 0, 0) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

2. (18 points) Consider the function $f(x, y, z) = \sin(\pi xyz)$ and the path $C: \mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + \mathbf{k}$, $-\infty \leq t \leq \infty$, both defined in space $\mathbb{R}^3$.

[6 pt] (a) Express the value of f along C as a function of t , and calculate its derivative at $t = 1$.

[6 pt] (b) Find the direction in which the value of f increases most rapidly at $(x, y, z) = (1, 2, 1)$.

[6 pt] (c) Calculate the derivative of f in the direction of the path C at $(x, y, z) = (1, 2, 1)$.

(a) $f(\mathbf{r}(t)) = \sin(2\pi t^2)$ [3 pt]

$$\left. \frac{d}{dt} \sin(2\pi t^2) \right|_{t=1} = 4\pi t \cos(2\pi t^2) \Big|_{t=1} = 4\pi. \quad [3 \text{ pt}]$$

(b) This is the direction of $\nabla f(1, 2, 1)$. [3 pt]

$$\begin{aligned} \nabla f(1, 2, 1) &= \left[\pi yz \cos(\pi xyz) \mathbf{i} \right. \\ &\quad \left. + \pi xz \cos(\pi xyz) \mathbf{j} \right. \\ &\quad \left. + \pi xy \cos(\pi xyz) \mathbf{k} \right] (1, 2, 1) \\ &= \pi (2\mathbf{i} + \mathbf{j} + 2\mathbf{k}) \quad [3 \text{ pt}] \end{aligned}$$

$\Rightarrow$ Direction of $\frac{1}{3} (2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.
(Normalization is optional.)

(c) $(1, 2, 1) = \mathbf{r}(1)$

$$\left. \frac{d\mathbf{r}}{dt}(1) \right| = \mathbf{i} + 2\mathbf{j}, \quad \left| \left. \frac{d\mathbf{r}}{dt}(1) \right| \right| = \sqrt{5}. \quad [3 \text{ pt}]$$

$\Rightarrow$ Directional derivative is

$$\frac{1}{|\dot{\mathbf{r}}(t)|} \left. \frac{d}{dt} f(\mathbf{r}(t)) \right|_{t=1} = \frac{4\pi}{\sqrt{5}} \quad [3 \text{ pt}]$$

or

$$\frac{\dot{\mathbf{r}}(1)}{|\dot{\mathbf{r}}(1)|} \cdot \nabla f(\mathbf{r}(1)) = \frac{4\pi}{\sqrt{5}} \quad [3 \text{ pt}]$$

3. (12 points) Consider the function $f(x, y) = e^x(x^2 - 2x - y^2)$ defined on the plane $\mathbb{R}^2$.

[6 pt] (a) Find all critical points of f .

[6 pt] (b) At each of the critical points found above, determine whether f has a local maximum, local minimum or saddle point.

(a) They are points where $f_x = f_y = 0$. [3 pt]

$$\begin{cases} f_x = e^x(x^2 - 2x - y^2) + e^x(2x - 2) \\ \quad = e^x(x^2 - y^2 - 2) \\ f_y = -2ye^x \end{cases}$$

$$f_y = 0 \Rightarrow y = 0$$

$$f_x = f_y = 0 \Rightarrow x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}.$$

Critical points are $(\pm\sqrt{2}, 0)$. [3 pt]

(b) We need to look at the signs of f_{xx} and $f_{xx}f_{yy} - f_{xy}^2$. [3 pt]

$$\begin{aligned} f_{xx} &= e^x(x^2 - y^2 - 2) + e^x \cdot 2x \\ &= e^x(x^2 - y^2 - 2 + 2x) \end{aligned}$$

$$f_{xy} = -2ye^x$$

$$f_{yy} = -2e^x$$

$$\begin{aligned} \text{At } (\sqrt{2}, 0), \quad f_{xx} > 0, \quad f_{xx}f_{yy} - f_{xy}^2 < 0 \\ \Rightarrow \text{saddle point} \end{aligned}$$

$$\begin{aligned} \text{At } (-\sqrt{2}, 0), \quad f_{xx} < 0, \quad f_{xx}f_{yy} - f_{xy}^2 > 0 \\ \Rightarrow \text{local maximum} \end{aligned}$$

[3 pt]

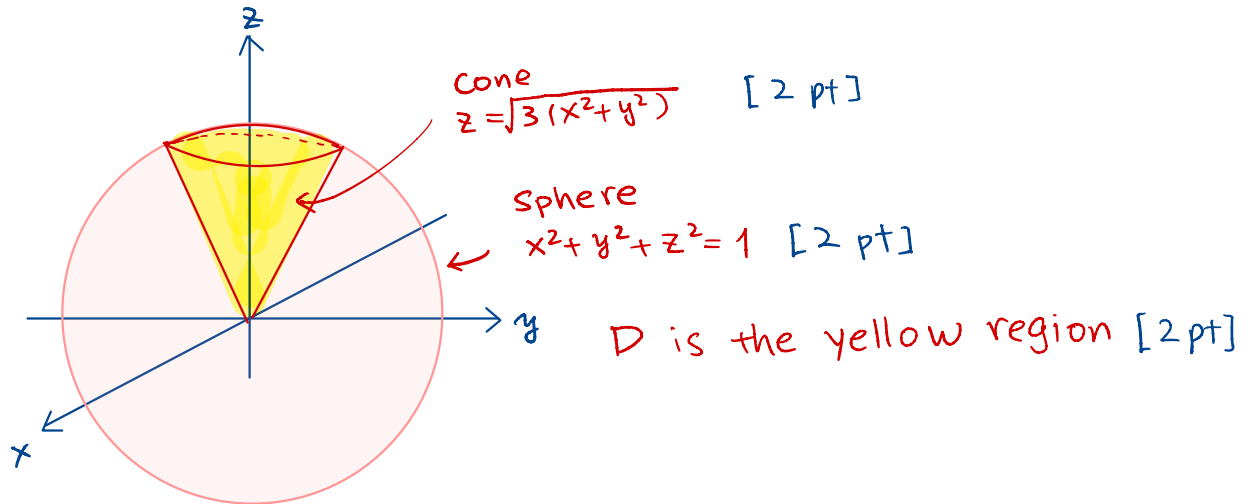
4. (16 points) Let D be the region in space $\mathbb{R}^3$ such that $\sqrt{3(x^2 + y^2)} \leq z \leq \sqrt{1 - x^2 - y^2}$.

[6 pt] (a) Sketch the region D . Make sure to include the x -, y - and z -axes.

[4 pt] (b) What values can each of the spherical coordinates ρ , ϕ , θ take in D ? Write down the inequalities the coordinates must satisfy.

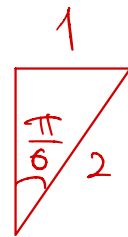
[6 pt] (c) Find the volume of D .

(a)



(b)
$$\left. \begin{aligned} 0 \leq \rho \leq 1 \\ 0 \leq \theta \leq 2\pi \end{aligned} \right\} [2 \text{ pt}]$$

$$0 \leq \phi \leq \frac{\pi}{6} [2 \text{ pt}]$$



(c)
$$\iiint_D dV = \int_0^{2\pi} \int_0^{\frac{\pi}{6}} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta [3 \text{ pt}]$$

$$= 2\pi \cdot \frac{1}{3} \cdot [-\cos \phi]_0^{\frac{\pi}{6}}$$

$$= \frac{2\pi}{3} \left(1 - \frac{\sqrt{3}}{2}\right). [3 \text{ pt}]$$

5. (14 points) Let $\mathbf{F}(x, y, z) = x^2 \sin(y)\mathbf{i} + 2x \cos(y)\mathbf{j} + 2\mathbf{k}$ be a vector field in space $\mathbb{R}^3$.

[4 pt] (a) Calculate the divergence $\nabla \cdot \mathbf{F}$.

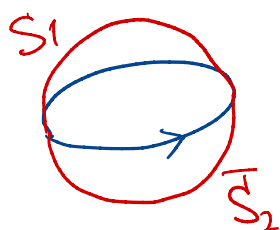
[4 pt] (b) If S_1 and S_2 are two oriented surfaces that share the same oriented boundary C , then $\iint_{S_1} \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iint_{S_2} \mathbf{F} \cdot \mathbf{n} \, d\sigma$. Explain why.

[6 pt] (c) Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma$, where S is the part of the unit sphere defined in the spherical coordinates by $\rho = 1$, $0 \leq \phi \leq \pi/3$, and $\mathbf{n}$ is the outward normal vector field.

$$(a) \quad \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (x^2 \sin y) + \frac{\partial}{\partial y} (2x \cos y) + \frac{\partial}{\partial z} 2 \quad [2 \text{ pt}]$$

$$= 2x \sin y - 2x \sin y = 0. \quad [2 \text{ pt}]$$

(b) We use the Divergence Theorem. [1 pt]



If $\bar{S}_2$ is the surface S_2 with opposite orientation, $S_1 \cup \bar{S}_2$ is a closed surface.

$$\iint_{S_1 \cup \bar{S}_2} \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV = 0$$

D
^
region enclosed by $S_1 \cup \bar{S}_2$.

Hence,

$$\iint_{S_1 \cup \bar{S}_2} \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iint_{S_1} \mathbf{F} \cdot \mathbf{n} \, d\sigma - \iint_{S_2} \mathbf{F} \cdot \mathbf{n} \, d\sigma = 0. \quad [3 \text{ pt}]$$

(c) By the result of (b), we can evaluate the integral for the disk S'
 $\{ 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, z = \cos \frac{\pi}{3} \}$ [3 pt]

Then, $\mathbf{n} = \mathbf{k}$ and

$$\iint \mathbf{F} \cdot \mathbf{n} \, d\sigma = 2 \cdot (\text{area of the disk}) = \frac{3\pi}{2}$$

[Attempts to evaluate the integral: 1-3 pts] [3 pt]